

Date	3 July 2026
Team ID	-
Project Name	Book-Store
Maximum Marks	4 Marks

PROJECT DESIGN PHASE: SOLUTION ARCHITECTURE

Solution Architecture Overview

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

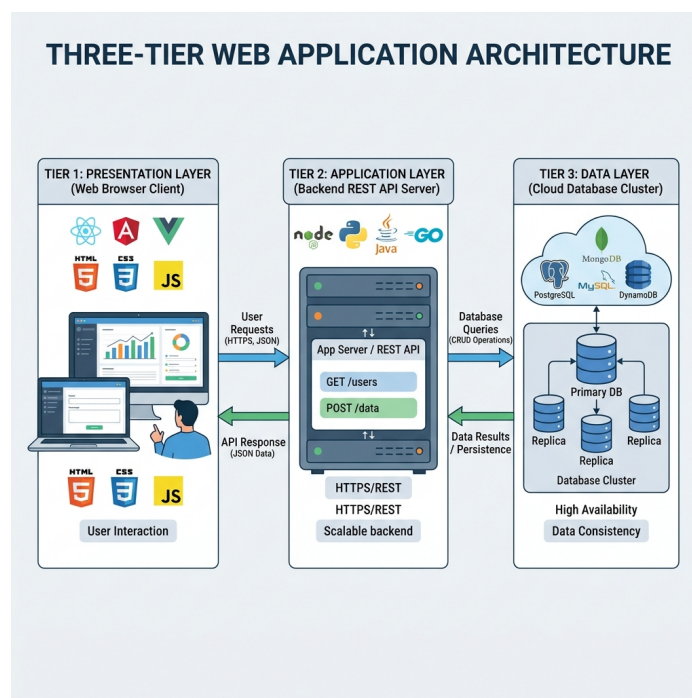


Figure 1: Solution Architecture Diagram detailing presentation, server application, and database storage tiers.

Architecture Layers Breakdown

1. Presentation Tier (Frontend Client)

A Single Page Application (SPA) built using React.js. It manages active routing paths via React Router, authenticates states via context APIs, and performs structured HTTP request queries to the application tier using Axios.

2. Application Tier (REST API Server)

An Express.js server hosted on Node.js. It registers endpoint routes for authorization, book inventory updates, and order generation. Middlewares verify incoming JWT tokens and manage local file directory storage via Multer.

3. Data Tier (Cloud Storage Cluster)

A cloud-hosted MongoDB Atlas replica set cluster. Object-document mapping is coordinated via Mongoose schema definitions (Admins, Sellers, Users, Books, Orders) which validate and update catalog records.
